

## SANSUI, China

WMO: 578320

Lat: 26.9609N

Lon: 108.6764E

Elev: 631

StdP: 93.97

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -2.9       | -1.6 | -7.0                          | 2.2 | 2.5  | -5.4 | 2.6 | 1.2  | 5.4                   | 5.7  | 4.7 | 5.1  | 1.5                   | 40   | 0.330 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.8                    | 33.2            | 23.5 | 32.2 | 23.4 | 31.1 | 23.2 | 25.2                | 30.0 | 24.7 | 29.4 | 24.2 | 28.7 | 2.6                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.0                            | 20.4 | 27.3 | 23.5 | 19.8 | 26.7 | 23.1 | 19.3 | 26.3 | 81.4          | 30.2 | 79.3 | 29.5 | 77.3 | 28.9 | 27.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 5.1               | 4.4  | 3.9 | -4.1                       | 35.0 | 1.0                | 1.0 | -4.8                                               | 35.8 | -5.4       | 36.4 | -5.9       | 36.9 | -6.7       | 37.7 |
|                   |      |     | -4.6                       | 26.0 | 1.1                | 0.8 | -5.4                                               | 26.5 | -6.0       | 27.0 | -6.6       | 27.4 | -7.3       | 28.0 |
|                   |      |     |                            | DB   |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     |                            | WB   |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 16.0   | 4.3   | 6.6   | 11.0  | 16.6  | 20.5  | 23.7  | 26.1  | 25.5  | 22.0  | 16.9  | 11.8  | 6.4  |
|                                                                                  | DBStd       | 8.22   | 3.77  | 4.54  | 4.55  | 4.27  | 3.07  | 2.54  | 1.93  | 2.28  | 3.25  | 3.36  | 3.65  | 3.06 |
|                                                                                  | HDD10.0     | 478    | 179   | 115   | 43    | 4     | 0     | 0     | 0     | 0     | 0     | 1     | 21    | 116  |
|                                                                                  | HDD18.3     | 1732   | 434   | 331   | 232   | 82    | 13    | 1     | 0     | 0     | 7     | 68    | 196   | 369  |
|                                                                                  | CDD10.0     | 2673   | 4     | 19    | 73    | 202   | 326   | 412   | 500   | 482   | 359   | 214   | 76    | 6    |
|                                                                                  | CDD18.3     | 883    | 0     | 1     | 4     | 29    | 81    | 162   | 242   | 223   | 117   | 23    | 2     | 0    |
|                                                                                  | CDH23.3     | 6840   | 0     | 7     | 38    | 236   | 542   | 1088  | 2153  | 1832  | 807   | 124   | 13    | 0    |
|                                                                                  | CDH26.7     | 2367   | 0     | 1     | 7     | 57    | 151   | 312   | 824   | 712   | 278   | 24    | 1     | 0    |
| Wind                                                                             | WSAvg       | 1.6    | 1.6   | 1.8   | 1.8   | 1.8   | 1.7   | 1.5   | 1.7   | 1.6   | 1.6   | 1.5   | 1.6   | 1.5  |
| Precipitation                                                                    | PrecAvg     | 1112   | 33    | 36    | 55    | 122   | 188   | 163   | 120   | 139   | 74    | 92    | 61    | 29   |
|                                                                                  | PrecMax     | 1548   | 80    | 92    | 102   | 220   | 367   | 304   | 311   | 284   | 166   | 204   | 212   | 76   |
|                                                                                  | PrecMin     | 854    | 8     | 9     | 17    | 51    | 80    | 55    | 23    | 31    | 1     | 22    | 2     | 0    |
|                                                                                  | PrecStd     | 162    | 15    | 20    | 21    | 48    | 67    | 74    | 65    | 64    | 42    | 46    | 45    | 19   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 18.7  | 23.7  | 27.4  | 30.7  | 31.9  | 32.5  | 34.6  | 34.7  | 33.5  | 29.3  | 25.3  | 18.5 |
|                                                                                  |             | MCWB   | 12.4  | 15.4  | 17.9  | 20.5  | 21.5  | 23.2  | 23.9  | 23.4  | 22.6  | 20.7  | 17.5  | 11.4 |
|                                                                                  | 2%          | DB     | 14.8  | 19.9  | 23.6  | 28.2  | 30.0  | 31.2  | 33.3  | 33.3  | 31.6  | 26.7  | 22.2  | 15.6 |
|                                                                                  |             | MCWB   | 9.8   | 13.7  | 16.9  | 20.1  | 21.0  | 23.3  | 24.0  | 23.6  | 22.5  | 19.1  | 15.9  | 9.9  |
|                                                                                  | 5%          | DB     | 12.0  | 16.1  | 20.5  | 26.1  | 28.4  | 29.8  | 32.1  | 32.0  | 30.0  | 24.3  | 19.8  | 13.2 |
|                                                                                  |             | MCWB   | 8.1   | 11.6  | 14.9  | 19.1  | 20.9  | 23.1  | 23.8  | 23.7  | 22.3  | 18.4  | 14.7  | 9.2  |
|                                                                                  | 10%         | DB     | 9.7   | 12.9  | 17.9  | 23.8  | 26.6  | 28.5  | 30.9  | 30.6  | 28.1  | 22.3  | 17.6  | 11.2 |
|                                                                                  |             | MCWB   | 6.9   | 9.7   | 13.9  | 18.4  | 20.5  | 22.8  | 23.8  | 23.5  | 21.6  | 17.7  | 14.0  | 8.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 13.5  | 16.0  | 19.1  | 22.3  | 24.1  | 25.2  | 26.1  | 25.8  | 24.5  | 22.1  | 18.9  | 13.2 |
|                                                                                  |             | MCDB   | 18.3  | 22.4  | 25.2  | 27.4  | 29.0  | 29.9  | 30.8  | 30.8  | 30.0  | 26.9  | 23.4  | 15.4 |
|                                                                                  | 2%          | WB     | 10.6  | 14.3  | 17.4  | 21.1  | 23.1  | 24.6  | 25.3  | 25.1  | 23.9  | 20.9  | 17.1  | 11.2 |
|                                                                                  |             | MCDB   | 13.4  | 19.3  | 22.4  | 26.3  | 27.7  | 28.9  | 30.0  | 30.1  | 29.2  | 24.3  | 20.5  | 14.2 |
|                                                                                  | 5%          | WB     | 8.8   | 12.2  | 16.0  | 20.1  | 22.3  | 24.1  | 24.8  | 24.6  | 23.2  | 19.7  | 15.9  | 9.8  |
|                                                                                  |             | MCDB   | 11.4  | 15.1  | 19.4  | 24.6  | 26.1  | 27.9  | 29.4  | 29.3  | 27.7  | 22.5  | 18.4  | 12.5 |
|                                                                                  | 10%         | WB     | 7.2   | 10.3  | 14.4  | 19.1  | 21.4  | 23.6  | 24.3  | 24.1  | 22.4  | 18.7  | 14.5  | 8.4  |
|                                                                                  |             | MCDB   | 9.3   | 12.6  | 17.0  | 22.9  | 24.8  | 27.0  | 28.8  | 28.5  | 26.5  | 21.2  | 16.9  | 10.8 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.2   | 5.9   | 6.9   | 7.7   | 7.7   | 6.7   | 7.8   | 8.2   | 8.1   | 6.7   | 6.9   | 6.2  |
|                                                                                  |             | MCDBR  | 11.8  | 13.3  | 13.5  | 13.0  | 12.5  | 9.7   | 10.8  | 11.3  | 11.9  | 11.4  | 12.1  | 12.7 |
|                                                                                  |             | MCWBR  | 6.8   | 7.0   | 6.3   | 5.3   | 4.5   | 3.0   | 2.8   | 3.0   | 3.7   | 4.4   | 5.8   | 7.1  |
|                                                                                  | 5% WB       | MCDBR  | 10.3  | 11.9  | 11.1  | 10.4  | 8.9   | 7.3   | 8.7   | 8.9   | 9.6   | 8.4   | 8.9   | 10.6 |
|                                                                                  |             | MCWBR  | 6.3   | 6.6   | 5.7   | 5.0   | 3.9   | 2.8   | 2.9   | 3.0   | 3.4   | 4.0   | 4.8   | 6.3  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.680  | 0.691 | 0.770 | 0.736 | 0.627 | 0.601 | 0.542 | 0.606 | 0.646 | 0.731 | 0.678 | 0.683 |      |
|                                                                                  | taud        | 1.478  | 1.492 | 1.404 | 1.504 | 1.759 | 1.845 | 2.017 | 1.821 | 1.697 | 1.482 | 1.533 | 1.485 |      |
|                                                                                  | Ebn at Noon | 559    | 605   | 592   | 635   | 711   | 725   | 768   | 717   | 670   | 568   | 557   | 530   |      |
|                                                                                  | Edh at Noon | 262    | 278   | 319   | 296   | 229   | 209   | 176   | 213   | 234   | 276   | 246   | 251   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.47   | 1.94  | 2.58  | 3.19  | 3.49  | 3.53  | 4.69  | 4.38  | 3.59  | 2.56  | 2.18  | 1.66  |      |
|                                                                                  | RadStd      | 0.36   | 0.48  | 0.35  | 0.50  | 0.58  | 0.51  | 0.41  | 0.38  | 0.44  | 0.38  | 0.44  | 0.37  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.31 | N/A     | N/A    | +0.79   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (8 neighbors) | +0.29 | N/A     | N/A    | +0.55   | N/A   | N/A   | N/A         | N/A     | +88     | N/A     |

Nomenclature: See separate page